



TITLE PAGE

- **Problem Statement ID** – 25004
- **Problem Statement Title**- Image based breed recognition for cattle and buffaloes of India
- **Theme**- Agriculture, Food Tech & Rural Development
- **PS Category**- Software
- **Team ID**-
- **Team Name**- Algo Ninjas



Image based breed recognition

1. Detailed Explanation of the Proposed Solution	2. How It Addresses the Problem	3. Innovation and Uniqueness of the Solution
• An AI-driven breed identification system that analysis the image and classified the breed • It uses the Convolutional Neural Networks (CNN) to extract feature from animal image • We will use a large labeled breed image database • The system provide real-time breed validation, that allow the FLW to accept the suggestion • This project is design to work even with low intermittent internet	• Eliminates Manual Judgment Errors • Standardized Breed Classification • Handles Complex Diversity • Reliable in Field Conditions • Supports Decision-Making • Integrates Seamlessly into BPA • Improves Data Quality	• AI-Driven Approach Instead of Manual Entry • Real-Time and On-Field Application • Robustness to Real-Life Variability • Extensive Breed Database Coverage • Low Technical Barrier for FLWs • Offline Mode Support • Scalable Integration

Technologies to Be Used

- Python
- JavaScript (React Native / ReactJS)
- SQL / NoSQL (e.g., PostgreSQL / MongoDB)

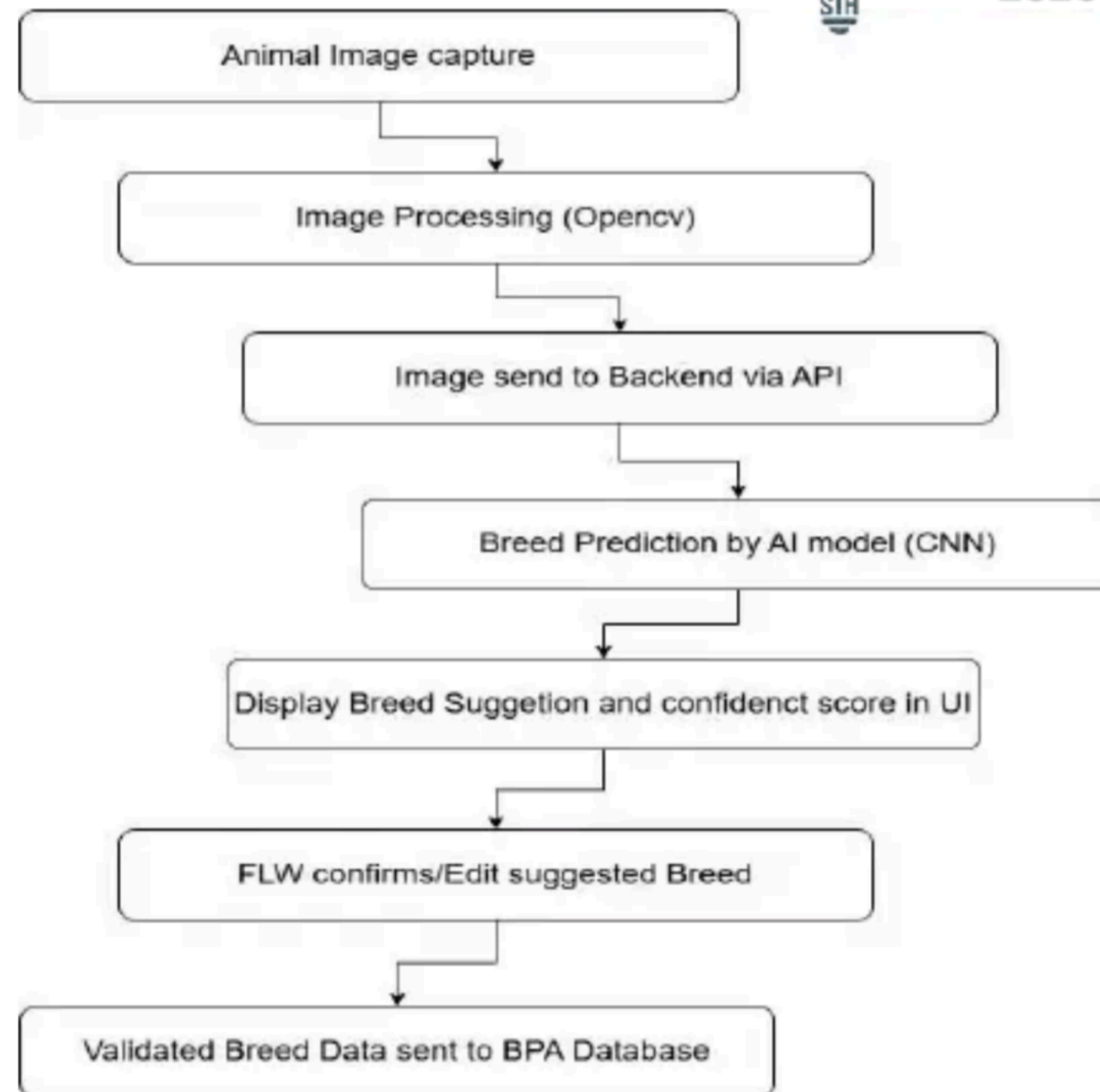
Methodology

- Model Development
- API Development
- Frontend Application (FLW Interface)
- Integration with BPA
- Testing and Validation

Frameworks and Libraries:

- TensorFlow / PyTorch
- OpenCV
- Flask / FastAPI
- React Native / ReactJS
- Docker

A L G O R I T H M



Analysis of the Feasibility of the Idea

- Proven image recognition technology (CNN models) is widely used in similar applications (e.g., face recognition, object classification).
- Minimal additional training is needed due to the simple, intuitive user interface.
- Cost of development is manageable compared to long-term benefits and Use of open-source frameworks (TensorFlow, OpenCV) reduces licensing costs.
- Farmers likely to adopt due to better accuracy and transparency in breed registration.

Potential Challenges and Risks

- Data Quality and Model Accuracy
- Hardware Constraints
- User Adoption Risk
- Integration Complexity
- Data Privacy and Security

Strategies for Overcoming Challenges

- Ensuring High Model Accuracy
- Supporting Offline Mode & LED
- Simplifying User Experience
- Smooth Integration
- Ensuring Data Security
- Monitoring and Feedback Loop

P R O T O T Y P E

HOW EXPLAINABLE AI WORKS IN OUR SOLUTION



- Horns
- Coat
- Hump

Breed: Sahiwal | Confidence: 92%

Potential Impact on the Target Audience (FLWs and Farmers)

1. Improves accuracy of breed registration for FLWs.
2. Reduces dependence on expert knowledge for breed identification.
3. Empowers FLWs with AI-driven decision support.
4. Speeds up the registration process.
5. Builds a reliable, standardized breed database.
6. Increases farmer confidence in livestock management services.
7. Enables real-time decision-making during registration.

Benefits of the Solution

1. Provides consistent and accurate breed classification.
2. Minimizes human errors and subjective judgments.
3. Saves time and effort for FLWs during data collection.
4. Requires minimal training to use the interface.
5. Supports sustainable and efficient livestock management.
6. Strengthens long-term planning for disease control and breeding programs.
7. Facilitates better resource utilization (feed, medicines).

These studies offer a strong foundation for our project, both in methodology and real-world applicability.

- [1]: https://eudl.eu/doi/10.4108/eai.23-11-2023.2343338?utm_source=chatgpt.com "Ensemble Learning Algorithm for Cattle Breed Identification using Computer Vision Techniques - EUDL"
- [2]: https://eudl.eu/doi/10.4108/eai.23-11-2023.2343289?utm_source=chatgpt.com "Mitigating Training Bias in Cattle Breed Identification through Orientation-Aware Framework - EUDL"
- [3]: https://researcher.manipal.edu/en/publications/computer-vision-based-approach-for-automatic-detection-of-dairy-cow-breed?utm_source=chatgpt.com "Computer Vision-Based Approach for Automatic Detection of Dairy Cow Breed - Manipal Academy of Higher Education, Manipal, India"
- [4]: https://www.mdpi.com/2079-9292/11/22/3791?utm_source=chatgpt.com "Computer Vision-Based Approach for Automatic Detection of Dairy Cow Breed"
- [5]: https://ui.adsabs.harvard.edu/abs/2024AIPC.3121d0004D/abstract?utm_source=chatgpt.com "A deep learning-based framework for identification of Sahiwal cow - ADS"
- [6]: https://krishikosh.egranth.ac.in/items/4ca5ec28-a558-406a-aca6-64449d724422?utm_source=chatgpt.com "INDIGENOUS CATTLE SAHIWAL AND RED SINDHI BREEDS IDENTIFICATION USING CONVOLUTIONAL NEURAL NETWORKS WITH TRANSFER LEARNING"
- [7]: https://www.propulsiontechjournal.com/index.php/journal/article/view/4296?utm_source=chatgpt.com "Cattle Breed Classification Techniques: Framework and Algorithm Evaluation | Tuijin Jishu/Journal of Propulsion Technology"